

COURSE TITLE : MODERN ARCHITECTURE
COURSE CODE : 6184
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER : 60
CREDITS : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
I	New materials	14
II	Development of new structures	16
III	Modern Architectural Concepts	16
IV	Modern Architecture in India	14
TOTAL		60

Course Objective :

Sl.	Sub	Student will be able to
1	1	Know about Industrial revolution and the development of new materials and techniques in building industry
	2	Understand the development of new structures
2	1	Learn about the modern movements in architecture and their expressions in buildings
	2	Identify the contributions of famous architects indulged in modern movements
3	1	Understand the architectural developments in modern India during the pre and post independence periods
	2	Understand the contributions of foreign architects in modern India towards city planning and building design

MODULE I: NEW MATERIALS

1.1 Outline the causes and effects of Industrial Revolution

1.1.1.Explain the causes of industrial revolution

1.1.2.Explain the effects of industrial revolution on transport, society and architecture

1.2 Comprehend the influence of new materials and techniques

1.2.1.Explain the role of concrete in the development of architecture

1.2.2.Explain the role of steel in the development of architecture

MODULE II: DEVELOPMENT OF NEW STRUCTURES

2.1. Understand the development of different types of Bridges

2.1.1. Know the development of timber, iron, steel and RCC bridges , suspension bridges- stiffened and un -stiffened.

2.1.2. Explain Iron Bridge, Coalbrookdale

2.2 Outline the structure of factory buildings

2.2.1 Identify the development of factory buildings

2.2.2. Explain AEG factory and Fagus Factory

2.3 Understand the importance of great exhibitions in developing new types of buildings

2.3.1. Explain the Great Exhibition London,1851- explain with sketches the features of Crystal Palace

2.3.2. Explain the Great Exhibition at Paris, 1889- explain with sketches the features of Eiffel Tower and Galerie des Machines

MODULE III : MODERN ARCHITECTURAL CONCEPTS

3.1 Outline the Influence of arts on architecture

3.1.1.Understand Arts and Crafts Movement, Art Nouveau, Expressionism

3.2 Identify the contributions of Antonio Gaudi

3.2.1. Explain the features of Casa Batllo and Casa Mila

3.3 Understand Organic Architecture

3.3.1.Identify the contributions of Frank Lloyd Wright towards organic architecture-

Falling Water, Guggenheim Museum

MODULE IV : MODERN ARCHITECTURE IN INDIA

4.1 Understand the architecture in India during the pre independence period

4.1.1. Know the period of transition 1900-1925 and Modern movement 1925- 1947

4.2 Understand architecture after independence

4.3 Understand the contributions of Foreign architects towards modern architecture in India

4.3.1 Understand the contributions of Edwin Lutyens – Design of New Delhi, India Gate, Rashtrapathi Bhavan

4.3.2 Understand the contributions of Le Corbusier- Design of Chandigarh City- Capitol complex

4.3.3 Understand the contributions of Louis Kahn – IIM Ahmedabad

4.3.4 Understand the contributions of Philip Johnson- National Centre for Performing Arts, Bombay

CONTENT DETAILS

MODULE I: NEW MATERIALS

Causes of industrial revolution and its effects on transportation, society and architecture- influence of new materials and techniques- role of concrete and steel in the development of architecture

MODULE II: DEVELOPMENT OF NEW STRUCTURES

Development of different types of Bridges - timber, iron, steel and RCC bridges , stiffened and unstiffened suspension bridges- Iron Bridge, Coalbrookdale- development of factory buildings - AEG factory - Fagus Factory- the Great Exhibition London,1851- sketches and features of Crystal Palace- the Great Exhibition at Paris, 1889- sketches and features of Eiffel Tower and Galerie des Machines

MODULE III: MODERN ARCHITECTURAL CONCEPTS

Influence of arts on architecture- Arts and Crafts Movement- Art Nouveau- Art Deco- Expressionism - contributions of Antonio Gaudi - Casa Batllo and Casa Mila - Organic Architecture - contributions of Frank Lloyd Wright towards organic architecture – Falling Water- Guggenheim Museum

MODULE IV: MODERN ARCHITECTURE IN INDIA

Architecture in India during the pre independence period- period of transition 1900-1925-Modern movement 1925- 1947- Architecture after independence - Contributions of Foreign architects - Edwin Lutyens – New Delhi –India Gate-Rashtrapathi Bhavan- Le Corbusier- Design of Chandigarh City- Capitol complex- Louis Kahn – IIM Ahmedabad - Philip Johnson- National Centre for Performing Arts, Bombay

REFERENCE

Architecture of the Twentieth Century	- Mary Hollingsworth
An Introduction to Twentieth Century Architecture	- Lucy Peel, Polly Powell, Alexander Garrett
Wonders of the World	- AA Publishing
Modern Architecture in India	- Sarbjith Bahga, Surinder Bahga, Yashinder Bahga
Building Beyond Borders	- Satish Grover
